

Government Officer Detailed Course

Price Statistics and Index Numbers

Level: Foundation to Intermediate

Audience: Government officers, statistical/data personnel

Format: Detailed training material with practical examples

■ Course overview

Price statistics measure changes in prices of specified goods and services. Government statistical systems use price information to monitor economic conditions, study inflation, support policy analysis and prepare statistical reports. Index numbers summarise price movements relative to a reference or base period. Reliable price statistics require consistent specifications, comparable observations and transparent weighting and methodology.

■ Learning objectives

- Explain price observations, product specifications, markets, periods and weights.
- Calculate simple price relatives and simple aggregate price indices.
- Understand weighted price indices.
- Explain the basic ideas behind Laspeyres, Paasche and Fisher indices.
- Interpret index values and distinguish an index level from an inflation rate.
- Understand year-on-year and month-on-month comparisons.
- Identify price-data quality problems such as missing observations, unusual prices and changes in product quality.

Module 1: Fundamentals of Price Statistics

- A price observation records the price of a specified item/service under defined conditions.
- Product specification should remain comparable across periods.
- Record the market, outlet or source, reference period and current period.
- A basket is a set of goods/services represented in an index.
- Weights reflect the relative importance of items in the basket.

Important terms: price, price quotation, specification, base period, current period, weight, basket, price relative.

Module summary: Consistent collection and comparable definitions are essential for meaningful price statistics.

Module 2: Index Numbers — Concepts and Formulas

- Price relative = $(\text{Current Price} \div \text{Base Price}) \times 100$.
- Simple aggregate price index = $(\text{Sum of current prices} \div \text{Sum of base prices}) \times 100$.
- Weighted indices give greater influence to items with greater weights.
- Laspeyres index uses base-period quantities: $L = [\Sigma(P_1Q_0) \div \Sigma(P_0Q_0)] \times 100$.
- Paasche index uses current-period quantities: $P = [\Sigma(P_1Q_1) \div \Sigma(P_0Q_1)] \times 100$.
- Fisher index = $\sqrt{(L \times P)}$.

Important terms: index number, price relative, simple aggregate, weighted index, Laspeyres, Paasche, Fisher.

Module summary: Index formulas differ mainly in how quantities/weights enter the calculation.

Module 3: Interpreting Index Values

- Index 100 means the same level as the reference base.
- Index 120 means the measured price level is 20% above the base-period level.
- Index 95 means the measured price level is 5% below the base-period level.
- An index level is not the same thing as the inflation rate for a particular period.
- Items with larger weights can have a larger influence on the overall index.
- An overall index can rise even when some individual item prices fall.

Important terms: base=100, index level, percentage change, inflation, weight, contribution.

Module summary: Interpretation must always specify the base/reference period and the relevant comparison.

Module 4: Inflation and Government Price Indicators

- Inflation refers to a sustained increase in the general price level as measured by an appropriate price indicator.
- Year-on-year compares a period with the corresponding period in the previous year.
- Month-on-month compares a period with the immediately preceding month.
- Consumer-oriented and wholesale-oriented indicators have different baskets, purposes and methodologies.
- Always check the basket, weights, reference period and methodology before interpreting an index.

Important terms: inflation, YoY, MoM, consumer prices, wholesale prices, basket, weight.

Module summary: Price indicators should be interpreted according to their population, basket, coverage and methodology.

Module 5: Price Data Quality and Practical Analysis

- Check for missing price observations.
- Investigate unusual price movements and possible data-entry errors.
- Maintain consistent units of measurement.
- Account for changes in product quality or specification using the approved methodology.
- Consider seasonal patterns where relevant.
- Document sources, collection methods, units and adjustments.

Important terms: missing observation, outlier, quality adjustment, unit, seasonality, documentation.

Module summary: Reliable indices depend on consistent collection, comparable specifications and transparent quality checks.

Module 6: Government Price Basket Case Study

- Step 1: Select representative items.
- Step 2: Record comparable prices for the reference and current periods.
- Step 3: Calculate item-level price relatives.
- Step 4: Apply approved weights.
- Step 5: Calculate and validate the overall index.
- Step 6: Interpret and report the result with the reference period and methodology.
- Step 7: Document limitations and unusual observations.

Important terms: basket, price relative, weighted index, contribution, reference period, methodology.

Module summary: Good price-statistics work moves from reliable observations to transparent calculations and careful interpretation.

Government / statistical example

Item	Base price (■)	Current price (■)	Price relative
Rice / kg	50	60	120
Pulses / kg	100	110	110
Cooking oil / litre	120	132	110
Illustrative fuel unit	100	125	125

Worked example / step-by-step explanation

Step 1: Rice base price = ■50 and current price = ■60.

Step 2: Price relative = $(60 \div 50) \times 100 = 120$.

Step 3: Interpretation: the rice price is 20% higher than its base-period price.

Important: An item-level price relative is not, by itself, a measure of economy-wide inflation.

Practical exercises

- Calculate the price relative for each item.
- Identify the item with the largest increase.
- Explain why an equal-weight average may not represent an actual household basket.
- Propose a logical basis for assigning weights.
- Explain the difference between an index level and an inflation rate.
- Write a short officer note interpreting the table.

20 AI-ready knowledge points

- Definition of price statistics.
- Price observation.
- Product specification.
- Base and current period.
- Price relative.
- Simple price-relative formula.
- Index number.
- Meaning of an index of 100.
- Meaning of an index of 120.
- Simple aggregate price index.
- Why weighted indices are used.
- Laspeyres and base quantities.
- Paasche and current quantities.
- Fisher index.
- Index level versus inflation rate.
- Year-on-year versus month-on-month.
- Price basket.
- Role of weights.
- Missing or unusual prices.
- Units, quality and documentation.

End-of-course review

- Explain price observation and price collection.
- Calculate a simple price relative.
- Explain the purpose of weights.
- Distinguish Laspeyres, Paasche and Fisher indices.

- Interpret an index relative to its base period.
- Distinguish an index level from an inflation rate.
- Identify price-data quality issues.
- Write a clear, evidence-based statistical note.